

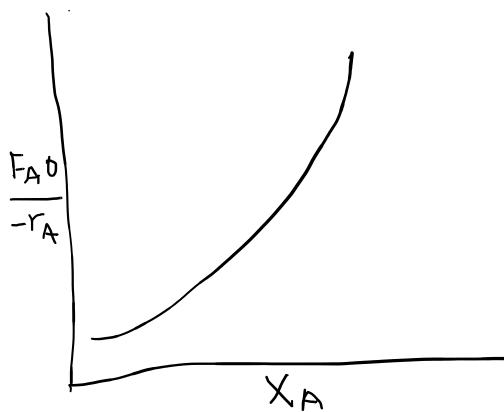
03 Levenspiel/Inverse Rate Plots: Graphical Illustrations of $V(X)$

Tuesday, May 12, 2020 12:04 PM

A Levenspiel Plot or Inverse Rate Plot plots the quantity $F_{A0}/-r_A$ versus X_A . These plots are used for visualizing the CSTR and PFR volumes needed to achieve a certain conversion for a reaction. They are also used for reactor sequencing.

Important note: These data in plots are based off of benchscale experiments; hence, these plots give *estimates* of necessary reactor volumes and appropriate sequences of reactors, but they *do not* substitute for carrying out the reactor design equations.

Example of a Levenspiel Plot:



$\uparrow X_A \downarrow C_A$

If $-r_1 = kC_1^n$, where $n > 0$,

$\downarrow C_A \downarrow kC_A^n$

In other words, the rate gets

smaller

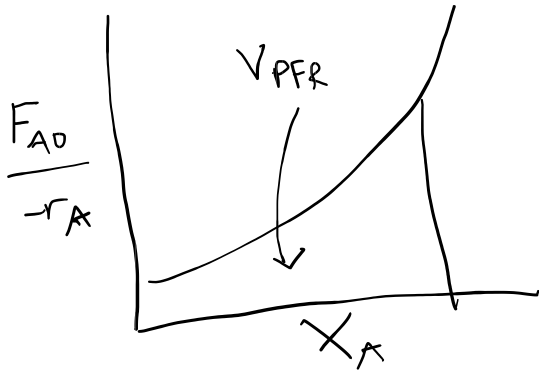
Hence, $\frac{F_{A0}}{-r_A}$ gets larger

How do we get the volume of a PFR using a Levenspiel Plot?

PFR mole balance: $\frac{dX_A}{dV} = \frac{-r_A}{F_{A0}}$, hence $V = F_{A0} \int \frac{dX_A}{-r_A}$

where r_A is a function of X_A

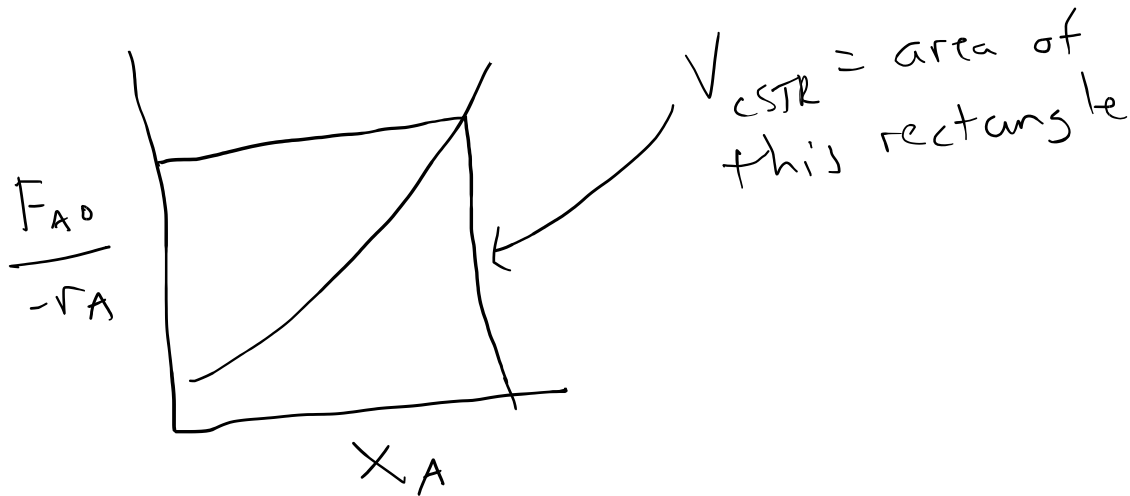
Hence, the PFR volume is the area under the Levenspiel curve.



How do we get the volume of a CSTR using a Levenspiel Plot?

CSTR mole balance: $V = \frac{F_{A0} X_A}{-r_A}$, hence $V = \left(\frac{F_{A0}}{-r_A} \right) X_A$

Hence, the CSTR volume is



If you have a PFR and a CSTR which are both the same Volume, which one will give the higher conversion for this reaction?

PFR

Why?

PFR maintains a higher concentration and thus a faster rate.

rate. The CSTR operates at the lowest possible concentration.

Let's practice! If your class is using the AIChE ConceptWarehouse, check to see if any problems about Levenspiel plots are open/assigned. We will discuss these in an upcoming synchronous session or class. A question to ponder: Why might $F_{A0}/-r_A$ decrease with X_A ?

We will come back to this when we discuss rate laws and the rate constant.